

Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 9: Multivariate Time Series & VAR

Duration: 60 minutes · No calculator · Answer on paper

Name: _____ Surname: _____

SCIPER (Student Card Number): _____

Exercise	Points
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2 .

Problem 1 (5 points)

We work with bivariate series $X_t = (X_t^{(1)}, X_t^{(2)})^\top$. Recall the matrix autocovariance $\Gamma_\tau = \text{Cov}(X_{t+\tau}, X_t)$ (*the lag τ sits in the first argument*), with (j, k) entry the cross-covariance $\gamma_\tau^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)})$, and the cross-correlation $\rho_\tau^{(j,k)} = \gamma_\tau^{(j,k)} / \sqrt{\gamma_0^{(j,j)} \gamma_0^{(k,k)}}$.

(a) State the symmetry relation linking $\Gamma_{-\tau}$ to Γ_τ , prove it from the definition, and explain in one sentence why the off-diagonal cross-covariance $\gamma_\tau^{(1,2)}$ need *not* be an even function of τ (unlike a scalar ACVS). (1.5 pts)

(b) Let $\{U_t^{(1)}\}$ and $\{U_t^{(2)}\}$ be two jointly stationary, mutually independent zero-mean series with $\text{Var}(U_t^{(j)}) = \sigma_j^2$, and let $\{\phi_t\}$ be a scalar mean-zero white noise of variance σ_ϕ^2 , independent of both. Define the common-shock model

$$Y_t^{(1)} = U_t^{(1)} + \phi_t, \quad Y_t^{(2)} = U_t^{(2)} + \phi_t.$$

Compute the cross-covariance $\gamma_\tau^{(1,2)}$ and the cross-correlation $\rho_\tau^{(1,2)}$ between $\{Y^{(1)}\}$ and $\{Y^{(2)}\}$ at *all* lags τ , and identify at which lag(s) the shared shock injects correlation. (2 pts)

(c) Specialise to $\sigma_1^2 = 3$, $\sigma_2^2 = 1$, $\sigma_\phi^2 = 1$ and report the numerical value of $\rho_0^{(1,2)}$ and of $\rho_\tau^{(1,2)}$ for $\tau \neq 0$. (1.5 pts)

Problem 2 (5 points)

(The delay process and its cross-spectrum.) Fix an integer delay ν and a gain $\alpha > 0$. Let $\{X_t^{(2)}\}$ be a zero-mean stationary series with autocovariance $\gamma_\tau^{(2,2)}$, spectral density $\mathcal{S}_2(f) = \sum_\tau \gamma_\tau^{(2,2)} e^{-2\pi i f \tau}$ and variance $\sigma_2^2 = \gamma_0^{(2,2)}$. Define a second channel as a scaled, delayed copy corrupted by noise,

$$X_t^{(1)} = \alpha X_{t-\nu}^{(2)} + \varepsilon_t, \quad t \in \mathbb{Z},$$

where $\{\varepsilon_t\}$ (variance σ^2) is white noise uncorrelated with $\{X_t^{(2)}\}$ at all lags.

(a) Show that the cross-covariance is a *shift* of the autocovariance of $X^{(2)}$, namely $\gamma_\tau^{(1,2)} = \alpha \gamma_{\tau-\nu}^{(2,2)}$, and deduce the cross-correlation $\rho_\tau^{(1,2)}$ at all lags in terms of $\rho^{(2,2)}$. (1.5 pts)

(b) Compute the cross-spectrum $S_{1,2}(f) = \sum_\tau \gamma_\tau^{(1,2)} e^{-2\pi i f \tau}$ in closed form, and read off its **amplitude** $|S_{1,2}(f)|$ and its **phase** $\arg S_{1,2}(f)$. Why is the phase *linear* in f , and what is its slope? (2 pts)

(c) Define the coherence $r_{1,2}(f) = S_{1,2}(f) / \sqrt{S_{1,1}(f) S_{2,2}(f)}$. Using $S_{1,1}(f) = \alpha^2 \mathcal{S}_2(f) + \sigma^2$ and $S_{2,2}(f) = \mathcal{S}_2(f)$, give $|r_{1,2}(f)|$, and show that in the *noiseless* limit $\sigma^2 \rightarrow 0$ the magnitude-squared coherence equals 1 at every frequency. (1.5 pts)

Problem 3 (5 points)

(VAR stability and the VAR(1) second-order structure.)

(a) Consider the bivariate VAR(2) process $X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \varepsilon_t$ with

$$\Phi_1 = \begin{pmatrix} 0.5 & 0 \\ 0.3 & 0.7 \end{pmatrix}, \quad \Phi_2 = \begin{pmatrix} 0 & 0 \\ 0.1 & -0.1 \end{pmatrix}.$$

Form the reverse characteristic polynomial $\det(I_2 - \Phi_1 z - \Phi_2 z^2)$, exploit the lower-triangular structure to *factor* it, find all its roots, and decide whether the VAR(2) is stable. State the companion-matrix (F) form of the stability test and verify it gives the same conclusion. (2.5 pts)

(b) Now take the *diagonal* VAR(1) process $X_t = \Phi X_{t-1} + \varepsilon_t$ with

$$\Phi = \text{diag}\left(\frac{1}{2}, \frac{1}{3}\right), \quad \{\varepsilon_t\} \sim \text{WN}(0, I_2).$$

Write its MA(∞) representation, and compute the stationary covariance Γ_0 and the lag-1 matrix Γ_1 in closed form (exact fractions). (1.5 pts)

(c) For the process of part (b), what is the coherence $r_{1,2}(f)$ between the two channels at every frequency, and why? (1 pt)

Problem 4 (5 points)

(Revision of Week 8: the tapered periodogram — bias kernel & unbiasedness.) Data $X_1, \dots, X_n$ ($T = \{1, \dots, n\}$, $\Delta = 1$) are observed from a zero-mean stationary process with spectral density $\mathcal{S}(f)$. A *data taper* is a sequence $\{h_t\}_{t \in T}$ (extended by $h_t = 0$ off T); the tapered periodogram is

$$\hat{\mathcal{S}}_h^{(p)}(f) = |J_h(f)|^2, \quad J_h(f) = \sum_{t \in T} h_t X_t e^{-2\pi i t f},$$

and $H(f) = \sum_t h_t e^{-2\pi i t f}$ denotes the discrete Fourier transform of the taper.

(a) Show that the expectation of the tapered periodogram is the frequency convolution

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df'.$$

(*Hint:* expand $|J_h|^2$ as a double sum, take expectations to get $\sum_\tau w_\tau \gamma_\tau e^{-2\pi i f \tau}$ with $w_\tau = \sum_t h_t h_{t+\tau}$, then identify w as an autocorrelation of the taper and apply the convolution theorem.) (2 pts)

(b) Suppose $\{X_t\}$ is white noise of variance σ^2 (so $\mathcal{S} \equiv \sigma^2$). Using part (a) and Parseval's relation $\int_{-1/2}^{1/2} |H(f')|^2 df' = \sum_t h_t^2$, prove that $\hat{\mathcal{S}}_h^{(p)}(f)$ is *unbiased* for $\mathcal{S}(f)$ at every f and every n **iff** $\|h\|_2^2 = \sum_t h_t^2 = 1$. (1.5 pts)

(c) Take $n = 3$ and the un-normalised triangular weights $g = (g_1, g_2, g_3) = (1, 2, 1)$. Find the constant c so that $h_t = c g_t$ is an admissible taper (i.e. $\|h\|_2^2 = 1$), and contrast, in one or two sentences, what tapering versus *multitapering* each fix about the raw periodogram. (1.5 pts)

